

GERHARDT ROETS

BSc Information Technology Student | Third Year | North-West University

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g3rrie.github.io/Portfolio



PROFESSIONAL SUMMARY

Motivated third-year Bachelor of Science in Information Technology student at North-West University, with hands-on academic experience in database design, artificial intelligence, and software development. Demonstrated ability to work collaboratively in group projects, including developing a relational database system and a C# first responder incident logging application. Proficient in SQL, Python, and C#, with practical exposure to GitHub and AI search algorithms. Eager to apply technical knowledge and problem-solving skills in a professional environment.

EDUCATION

Bachelor of Science in Information Technology

2024 – Present

North-West University — Potchefstroom Campus | Field of Study: Information Systems

Relevant Modules:

- CMPG 213 – System Analysis and Design
- CMPG 221 – Data Structures and Algorithms
- CMPG 311 – Database Management Systems
- CMPG 312 – Decision Support Systems
- CMPG 313 – Artificial Intelligence
- CMPG 315 – Computer Networks & Network Security

Distinctions achieved:

- Apps & Advanced User Interface Programming (CMPG 212) — 80%
- Data Structures & Algorithms (CMPG 221) — 76%
- System Analysis & Design 2 (CMPG 223) — 75%

RESEARCH PROJECTS

System Analysis & Design — First Responder Incident Logging System

2nd Year

- Developed a First Responder Incident Logging System in C# (Windows Forms) as part of a group project, enabling emergency personnel to log and track on-scene incidents in real time.
- Utilised GitHub for version control and collaborative development — first professional exposure to Git workflows and repository management.

Database Management Systems — Stoep & Stay Properties

3rd Year

- Designed and developed a relational database for a fictional real estate agency, producing a fully normalised 19-table schema using ERD modelling (crow's foot notation) and SQL.
- Applied 1NF through 3NF normalisation principles, eliminating partial and transitive dependencies across all tables.

Artificial Intelligence — Healthcare Diagnostic Chatbot

3rd Year

- Developed a functional healthcare diagnostic chatbot in Python using First-Order Logic (FOL), debugging and resolving eleven logic errors in the inference engine.
- Implemented and analysed graph-based search algorithms including A* Search and Greedy Best-First Search on real-world graph problems.

TECHNICAL SKILLS

■ SQL & Oracle Database	■ Python
■ C# (Windows Forms)	■ Java
■ GitHub & Version Control	■ ERD Design & Data Modelling
■ Database Normalisation (1NF–3NF)	■ AI Search Algorithms (A*, Greedy BFS)
■ Network Configuration (VLANs, DHCP)	■ Linear Programming & Optimisation

SOFT SKILLS

■ Teamwork & Collaboration	■ Problem-solving
■ Attention to Detail	■ Self-directed Learning
■ Friendly, Positive Attitude	■ Adaptability

LANGUAGES

English — Fluent (read, write, speak)

Afrikaans — Native (read, write, speak)

REFERENCES

Available on request.